TIPE – Eddy Current Braking

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Session 2025

Outline

Introduction

1. Theoretical model
   - Model construction
   - Using the model
2. Experimental model
3. Experimental results
   - Results and uncertainties
   - Critique of the model and sources of uncertainty

Conclusion

Appendices

A Brief Overview of Eddy Currents

Definition

Eddy currents are induced electric currents flowing in conducting bodies

- Discovered by Léon Foucault in 1851
- Direct consequence of electromagnetic induction

Many applications:

- Non-destructive testing
- Brazing
- Separator for sorting materials

Start of the project

Operating principle of the electromagnetic brake:
Lenz's law $\Longrightarrow$ the Laplace forces due to the induced currents oppose the initial motion.

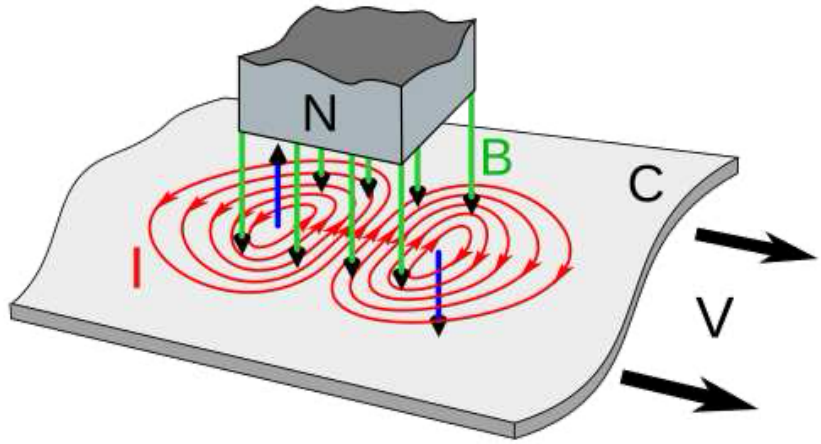


Figure: Eddy currents generated under a moving magnet[a]

[a] fr.wikipedia.org/wiki/Courants_de_Foucault

Problem statement and solution steps

Problem statement

Which characteristics of an eddy current retarder make it possible to modify the braking capacity of the system and its performance in road freight transport?

Solution steps

1. Theoretical model
2. Construction of the experimental model
3. Analysis of the results

1. Theoretical model

Model construction

Using the Laplace forces and computing the magnetic power

$$\vec{F_L} = q(\vec{E} + \vec{v} \wedge \vec{B})$$

Lorentz force

$$\mathrm{d}\vec{F_L} = q(r\omega\vec{e_\theta} \wedge B\vec{e_z})$$

$$\iff \vec{F_{Lm}} = qr\omega B\vec{e_r}$$

$$\Rightarrow \vec{E_{induced}} = \vec{v} \wedge \vec{B} = \omega \cdot r \cdot B\vec{e_r}$$

We assume that the electric field is constant, so we only consider the magnetic field

Model construction

Calculation of the power dissipated by the system

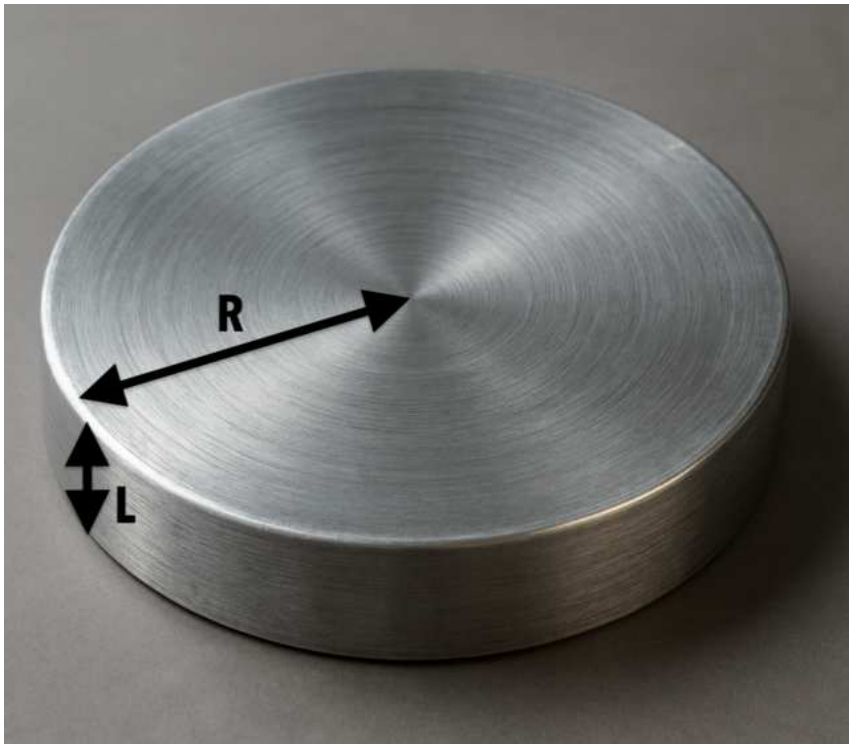
Local Ohm's law: $\vec{j}(r) = \frac{1}{\rho}\vec{E}(r)$

Power: $p(r) = \frac{1}{\rho}\vec{E}(r)^2$

$$\iff p(r) = \frac{B^2 \cdot w^2 \cdot r^2}{\rho}$$

$$P_{Total} = \int_0^R p(r) \cdot d\tau$$

$$P_{Total} = \frac{\pi B^2 \omega^2 R^4}{2\rho}$$



Disc of resistivity ρ [a]

[a] AI-generated photo

Power dissipated by the system

Power balance

$$P_{Total} = \frac{\pi B^2 \omega^2 R^4}{k\rho}$$

where $k \in [2; 8]$

- $P_{Totale} \rightarrow$ power that the retarder can theoretically dissipate
- k $\rightarrow$ correction factor taking the geometry into account[a]

[a] Result from the literature:
Fausto Fiorillo, *Measurement and Characterization of Magnetic Materials* (2004)

Model prediction

$$P_{\text{to dissipate}} = C \cdot \omega = C \cdot \frac{60}{2\pi} \cdot N$$

According to the manufacturer's data:

$$C_{nominal} = 0.39Nm$$

Therefore

$$P_{\text{to dissipate}} = 22 \cdot 10^3 W$$

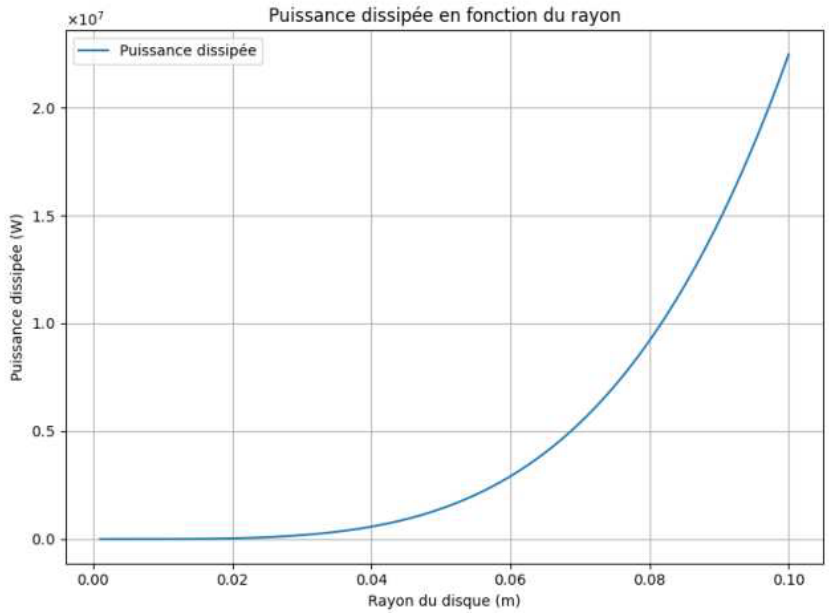


Figure: Estimates of the dissipated power as a function of R

Model prediction

Remark

The values seem high, but they are only valid for $N_{nominal} = 6000\ \mathrm{rpm}$ and $B = 1T$ ($B_{measured} = 0.58T$)

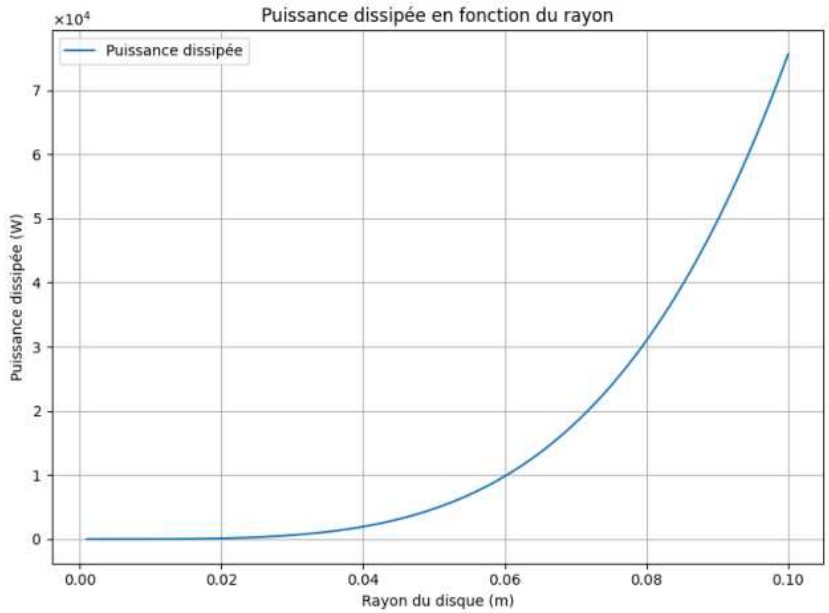


Figure: Revised estimates of the dissipated power as a function of R

Quantifying the heating of the disc

$$P_{\text{total}} = \frac{\pi B^2 \omega^2 R^4}{k\rho}$$

$$\Rightarrow P_{\text{total}} = \frac{\pi B^2 \left(\frac{2\pi}{60} \times N\right)^2 R^4}{k\rho}$$

$$\Rightarrow P_{\text{total}} = \alpha \times N^2$$

with:

$$\alpha = \frac{B^2 \pi^3 R^4}{900 \cdot k\rho}$$

Thermal study

Heat balance

- $\delta Q_{Foucault} = \alpha \times N^2 \cdot dt$
- $\delta Q_{air} = -h \cdot S(T(t) - T_{air})dt$ (Newton's law of cooling)

$$C\frac{dT(t)}{dt} = \alpha N^2 - h \cdot S(T(t) - T_{air})$$

$$\frac{dT(t)}{dt} + \frac{hS}{C}T(t) = \frac{\alpha}{C}N^2 + \frac{h \cdot S}{C}T_{air}$$

$$\Rightarrow T(t, N) = \frac{\alpha}{C}N^2(1 - e^{\frac{-t}{\tau}}) + T_{air}$$

Predicting the dependence of ρ on temperature

Using the law:

$$\rho(T) = \rho_0(1 + \alpha_0(T - T_0))^a$$

Where:

- $\rho_0 \rightarrow$ resistivity at T_0
- $\alpha_0 \rightarrow$ temperature coefficient at T_0
- $T_0 \rightarrow$ reference temperature (taken equal to 20 degrees)

[a] Ward, M. R. *Electric circuits and machines* (1971)

Theoretical results

Variation of the resistivity

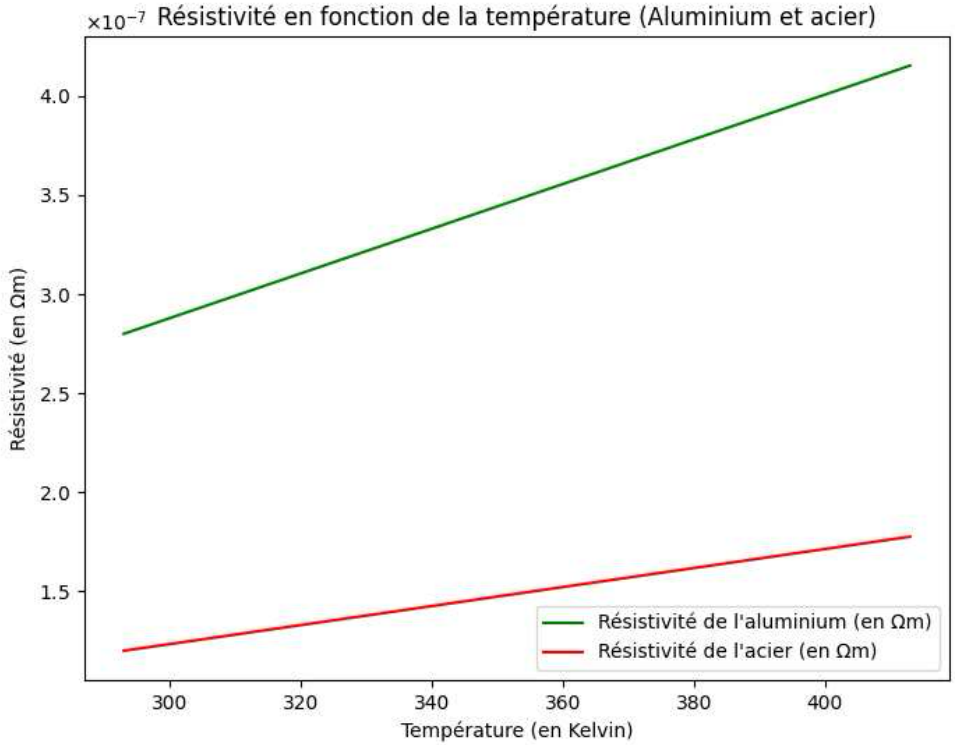


Figure: Variation of the resistivity

Values used for the model [a]

- For aluminium
  - $\alpha_0 = 4.0 \times 10^{-3} \mathrm{K}^{-1}$
  - $\rho_0 = 2.8 \times 10^{-8} \Omega\mathrm{m}$
- For steel
  - $\alpha_0 = 4.0 \times 10^{-3} \mathrm{K}^{-1}$
  - $\rho_0 = 1.2 \times 10^{-7} \Omega\mathrm{m}$

[a] Source of the values used: Wikipedia

Theoretical results

Variation of the resistivity

$$\rho_{aluminium}(T = 423K) = 4.15 \times 10^{-7}\Omega\text{m}$$

$$\rho_{steel}(T = 423K) = 1.7759 \times 10^{-7}\Omega\text{m}$$

$$\Rightarrow \Delta\rho_{steel} = 5.8 \times 10^{-8}\Omega\text{m} << \rho_{steel}(293K)$$

Conclusion of the study

$\Rightarrow$ Variations of ρ are negligible for steel in our study

Theoretical results

Evolution of the temperature

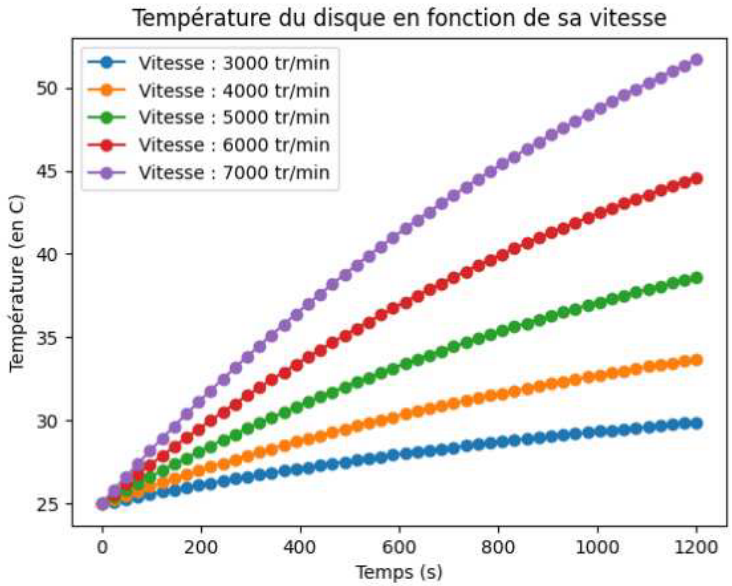


Figure: Simulation of the heating as a function of the shaft rotation speed (over 20 minutes)

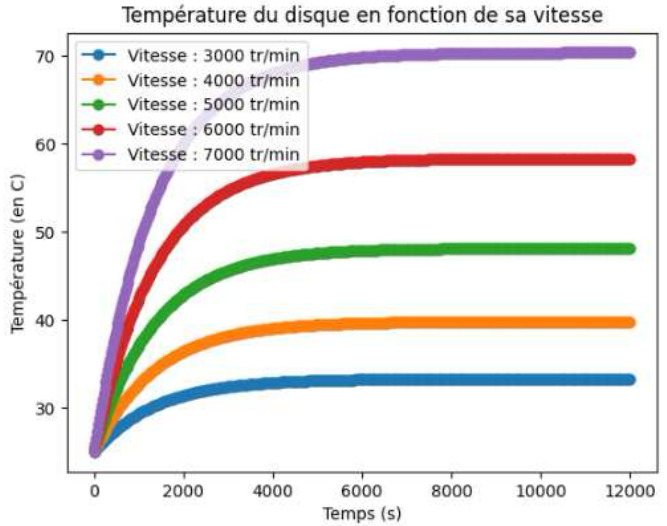


Figure: Simulation of the heating as a function of the shaft rotation speed (over 3 h)

$$T(t, N) = \frac{\alpha}{C} N^2 (1 - e^{\frac{-t}{\tau}}) + T_{air}$$

2. Experimental model

Experiments to carry out:

1. Link between resistivity and temperature
2. Braking capacity of the scale model

Protocol no. 1

Objective

Experimentally measure the variation of resistivity as a function of temperature

Method used:
Four-point method
According to the local Ohm's law:

$$R = \rho \frac{e}{A}$$

- e is the thickness of the disc
- A is the cross-sectional area

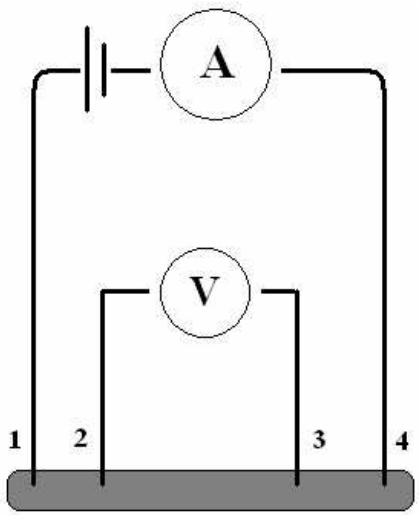


Figure: Principle of the four-point method

Protocol no. 1

Method used:
Four-point method
According to the local Ohm's law:

$$\rho = \frac{U}{I} \cdot \frac{A}{e}$$

- e is the thickness of the disc
- A is the cross-sectional area

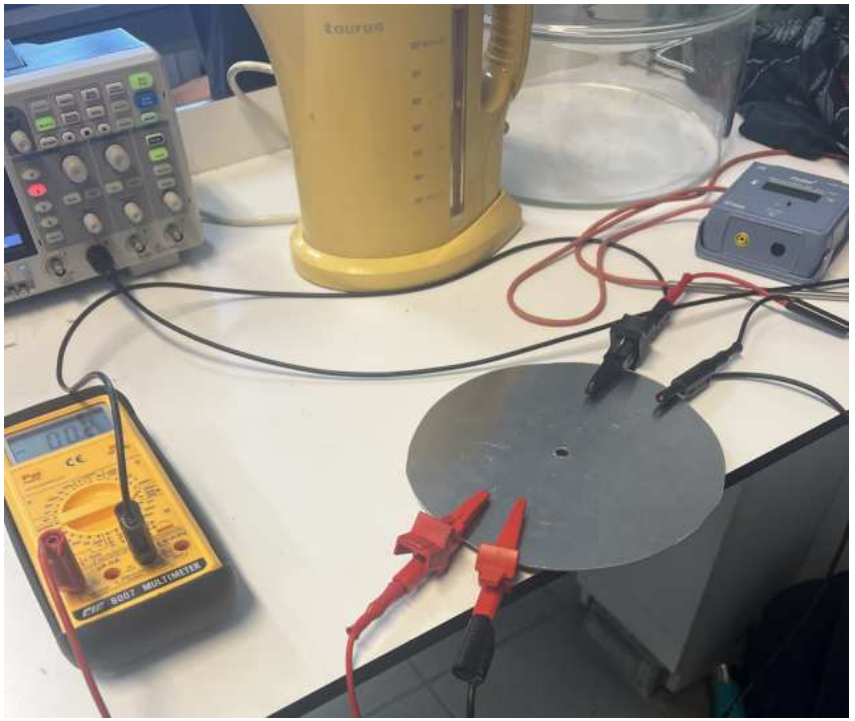


Figure: Experimental setup

Protocol no. 2

Objective

Measure the intensity of the eddy currents

- Connect the coils in series to a data acquisition card
- After the disc is set in rotation, automatic measurements are taken with a time step of 0.003 s



Figure: Experimental setup

Protocol no. 3

Objective

Quantify the energy dissipation (through the variation of the disc temperature)

- Supply the coils (in series) with a sinusoidal voltage
- Measure the disc temperature at regular intervals

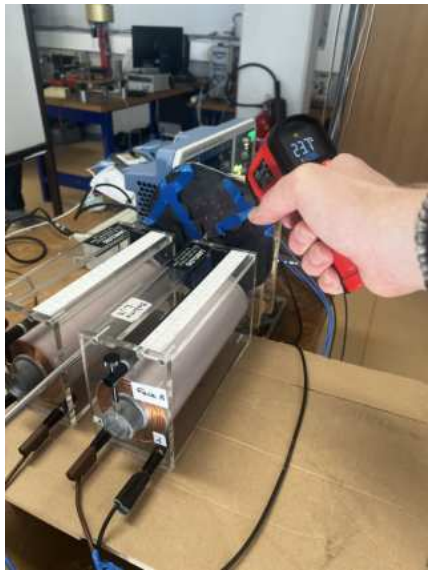


Figure: Experimental setup

3. Experimental results

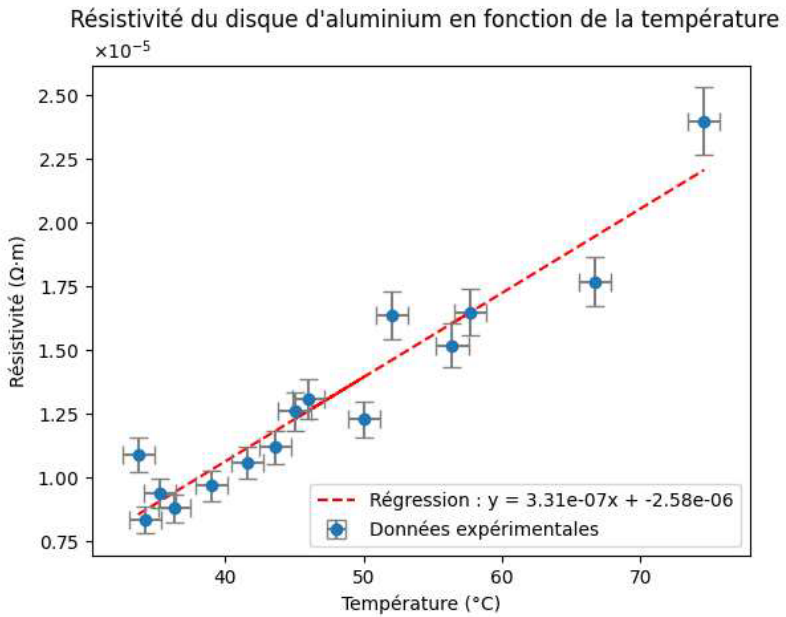
Evaluation of measurement uncertainties

Use of Type B uncertainty

Results

First experiment – variation of resistivity

Results for an aluminium disc



Results for a steel disc

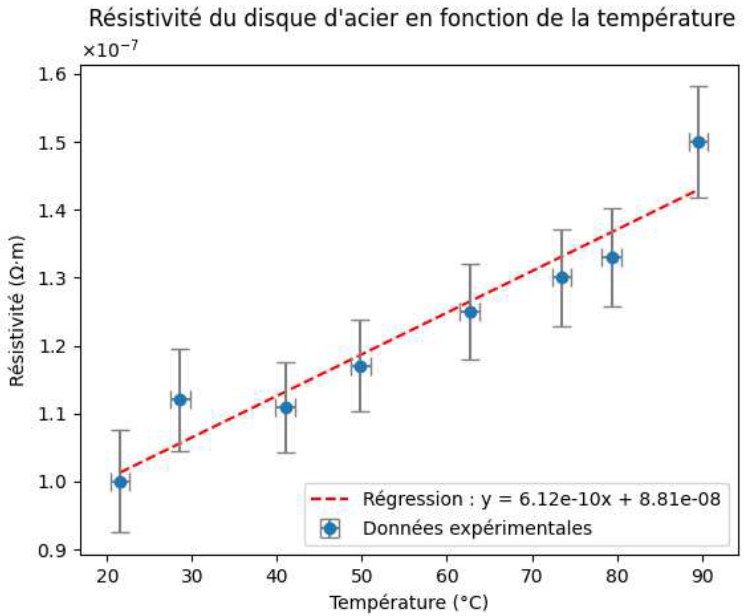


Figure: Results with the steel disc

Results

Comparison with the model

Results for the aluminium disc

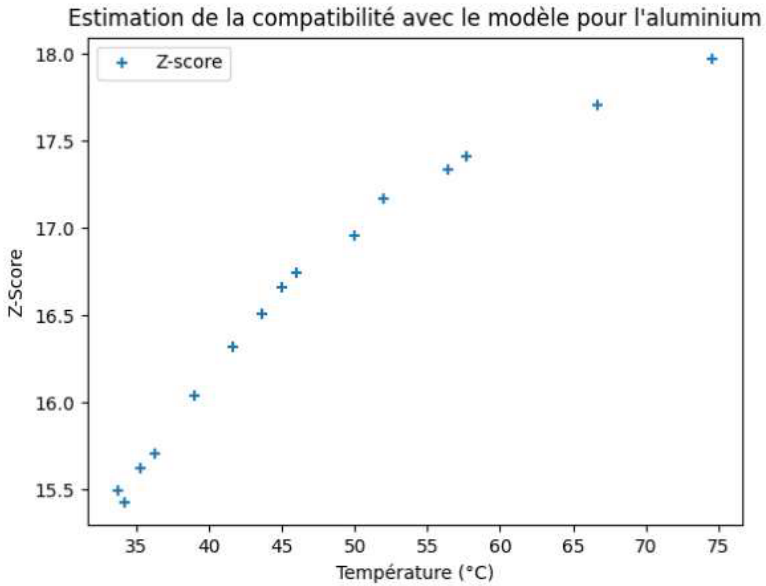


Figure: Results with the aluminium disc

Results for a steel disc

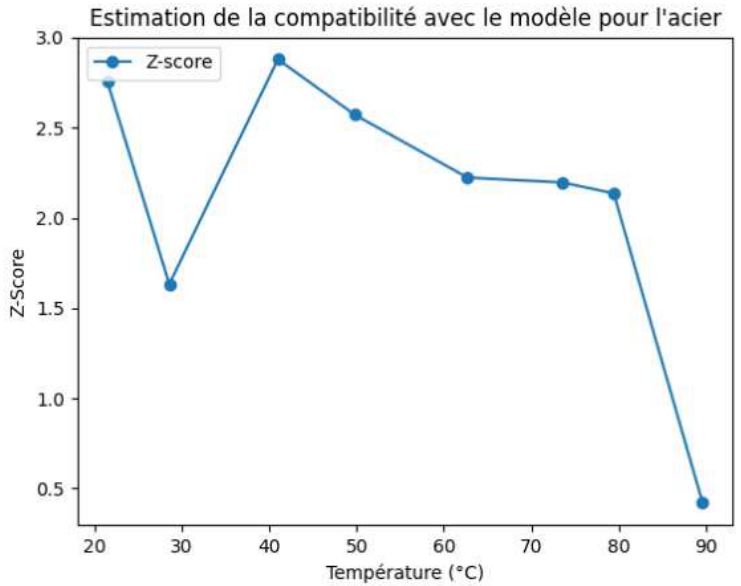


Figure: Results with the steel disc

Results

Second experiment – Eddy current induced in the coils

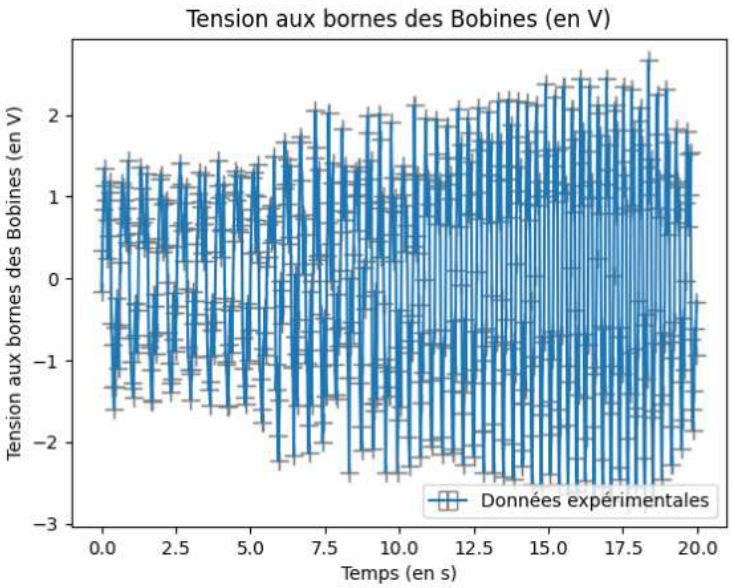


Figure: Voltage across the coils

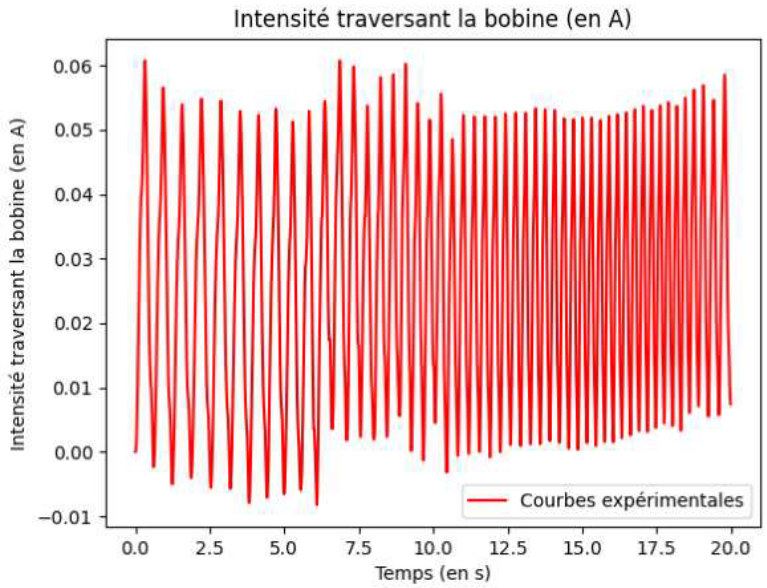


Figure: Current intensity in the coils

Results

Third experiment – Quantifying the heating of the disc

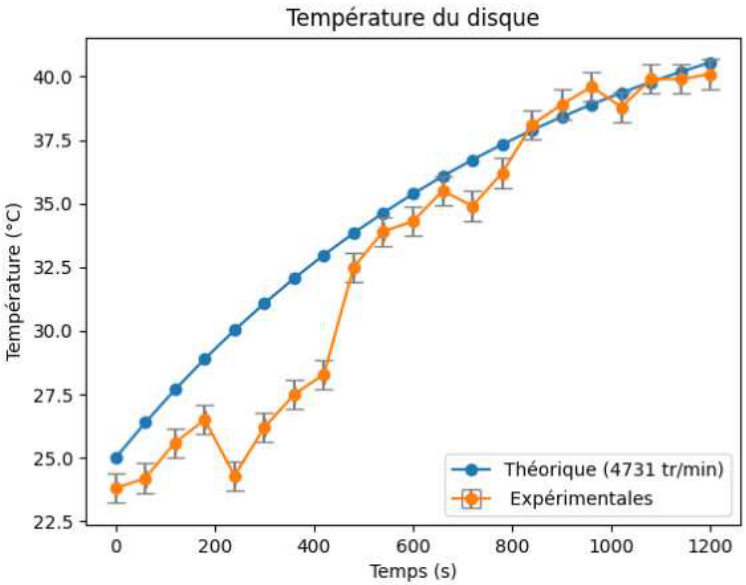


Figure: Theoretical model and experimental results

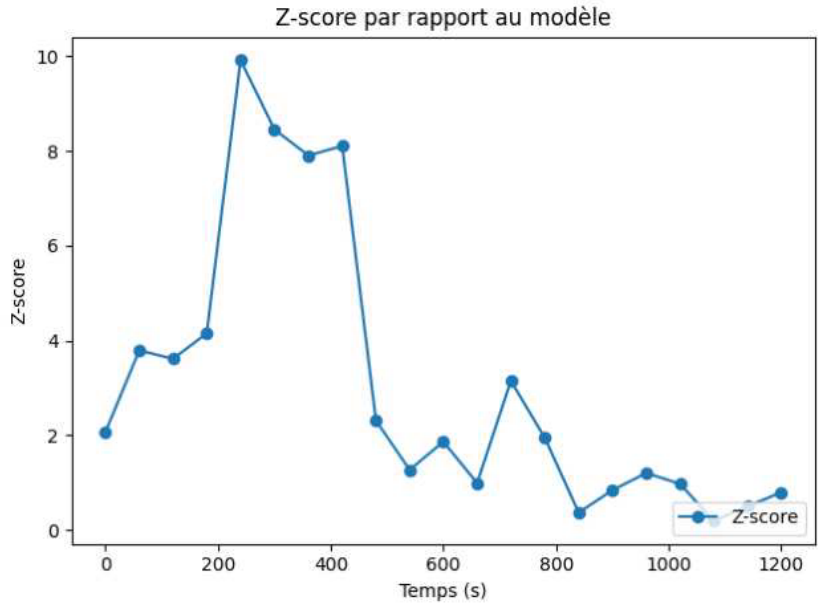
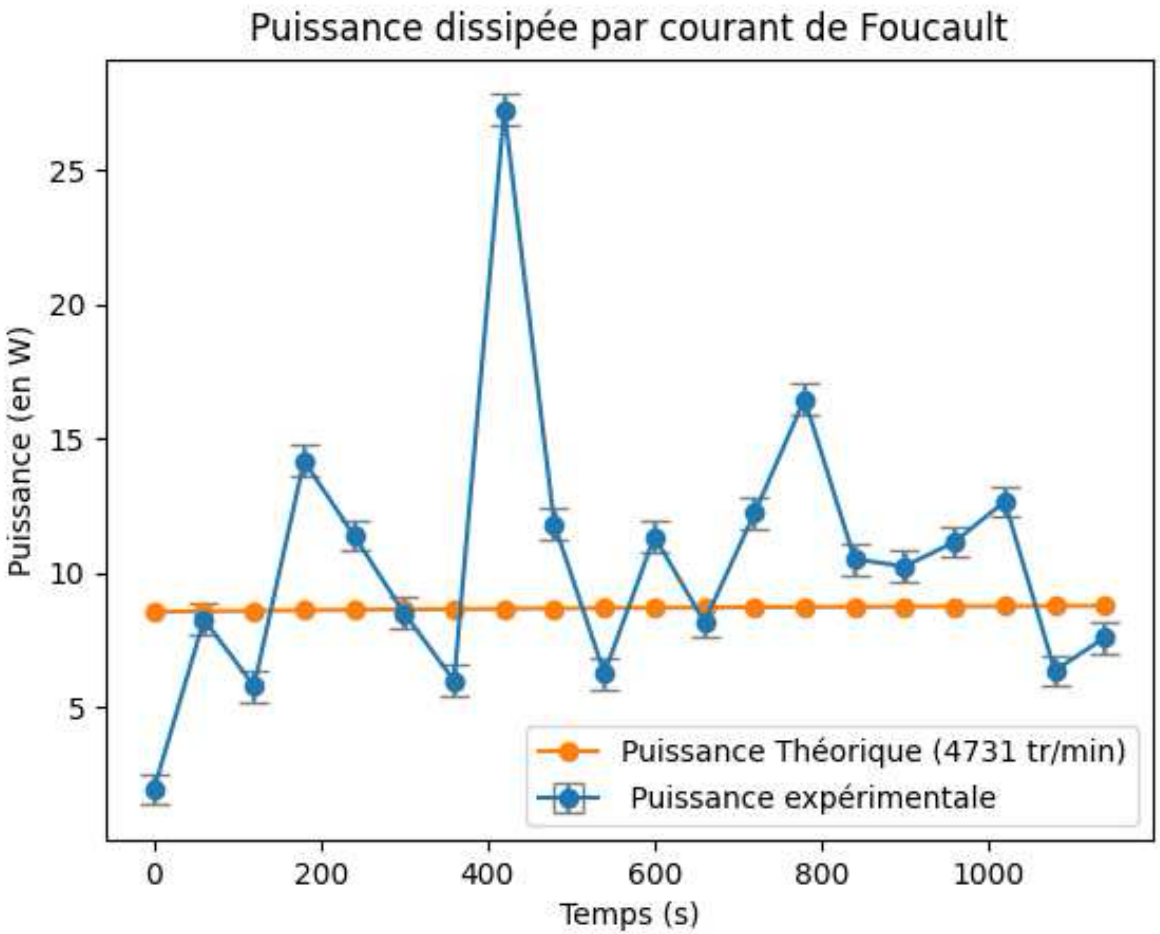


Figure: Evolution of the Z-score over time

Results

Third experiment – Quantifying the power dissipated by the disc



Sources of uncertainty and errors in my model and in the experimental setup

1. For the model
   - Values taken from tables that did not correspond exactly to the material
   - Model based on the assumption of a uniform magnetic field
2. For the experimental setup
   - Low forces transmitted by the scale model (unlike the system fitted to heavy goods vehicles)
   - Temperature measurements not always taken at the same location (variation of the order of a centimetre)

Summary

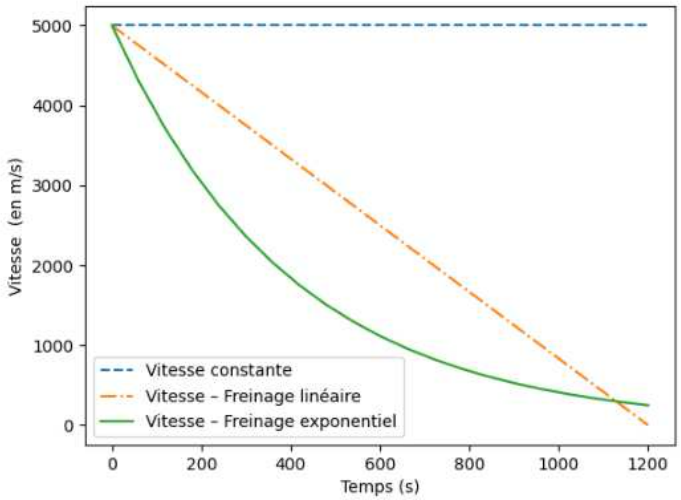


Figure: Speed profile used to compare braking

Factors influencing braking

- Disc geometry
- Initial speed (here 5000 rpm)

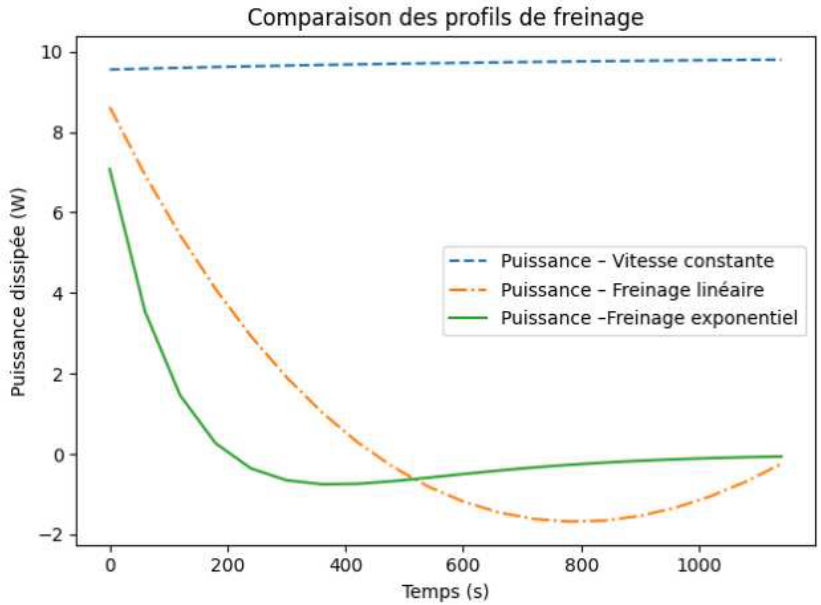


Figure: Dissipated power as a function of the variation of speed

Ideas for improving and continuing the study

- Use several discs in series
- Replace the steel disc with a copper disc
- Carry out a numerical simulation with more accurate software
  $\rightarrow$ for better modelling of the current distribution
- Try to perform braking with different profiles as modelled

Thank you for your attention!

Characteristics of the motor used

895 Dual Ball Bearing DC Motor

Double Ball Bearing DC motor with built-in cooling fan. High torque with wide operating voltage 6~20Vdc. Suitable for motor tools application and DIY projects.



SKU: FAM1023

Specifications:

- Motor Type: 895.
- Operating Voltage: 6~20Vdc. (Nominal 12Vdc)
- No Load Speed: 6,000 RPM @ 12V.
- Rated current: 1.0A @ 12V.
- Mechanical Output Power: 360W.
- Rated Torque: 39N.cm/4Kg.cm.
- Cooling Fan: Internal
- Overall Size: 102 x Ø50mm.
- Shaft: Full Round Type Ø5mm.
- Mounting Screw Size: 29mm (M4) & 32mm (M5).
- Weight: 510g.

Figure: Characteristics of the scale model's motor

Python code used to build the theoretical model

```python
import matplotlib.pyplot as plt
import math
import numpy as np
from matplotlib.ticker import ScalarFormatter

# Paramètres physiques
B = 58*10**(-3)  # Tesla
rho = 69*10**-8  # ohm·m
N = 6000  # tours/min
omega = 2 * math.pi * N / 60  # rad/s

# Rayon du disque (de 1 mm à 0.5 m)
X = np.linspace(0.001, 0.1, 500)
Puissance_dissipée = []

# Calcul puissance dissipée
for R in X:
    P = (math.pi * B**2 * omega**2 * R**4 ) / (8 * rho)
    Puissance_dissipée.append(P)

# Graphique
fig, ax = plt.subplots(figsize=(8, 6))
ax.plot(X, Puissance_dissipée, label="Puissance dissipée")

# Ecriture scientifique
formatter = ScalarFormatter(useMathText=True)
formatter.set_scientific(True)
formatter.set_powerlimits((-2, 2))
ax.yaxis.set_major_formatter(formatter)

# Graphique
ax.set_xlabel("Rayon du disque (m)")
ax.set_ylabel("Puissance dissipée (W)")
ax.set_title("Puissance dissipée en fonction du rayon")
ax.legend()
ax.grid(True)
plt.tight_layout()
plt.savefig("Puissance dissipee2")
plt.show()
```

Python code used to process the results

```python
import matplotlib.pyplot as plt
import numpy as np
from scipy.stats import linregress
from matplotlib.ticker import ScalarFormatter
from math import sqrt

X = np.array([33.7,34.2,35.3,36.3,39.0,41.6,43.6,46.0,50.0,45.0,52.0,57.7,56.4,66.7,74.6])
Y = np.array([1.09e-5, 8.33e-6, 9.39e-6, 8.81e-6, 9.68e-6, 1.06e-5, 1.12e-5, 1.31e-5, 1.23e-5,
              1.26e-5, 1.64e-5, 1.65e-5, 1.52e-5, 1.77e-5, 2.40e-5])
Xerr = np.array([2/sqrt(3) for _ in range(len(X))])
Yerr = np.array([Y[i]*sqrt((Xerr[i]/X[i])**2+(0.1/1.9)**2) for i in range(len(Y))])

# Régression linéaire
slope, intercept, r_value, p_value, std_err = linregress(X, Y)
y_fit = slope * X + intercept

def Traitement_brut():
        # Création de la figure et des axes
        fig, ax = plt.subplots()

        # Affichage de la régression
        ax.plot(X, y_fit, color='red', linestyle='--', label=f"Régression : y = {slope:.2e}x + {intercept:.2e}")

        # Affichage des points expérimentaux avec barres d'erreur
        ax.errorbar(X, Y, yerr=Yerr, xerr=Xerr, fmt='o', ecolor='gray', capsize=5, label="Données expérimentales")

        # Format scientifique pour les ordonnées
        formatter = ScalarFormatter(useMathText=True)
        formatter.set_scientific(True)
        formatter.set_powerlimits((-3, 3))
        ax.yaxis.set_major_formatter(formatter)

        # Ajout des titres et labels
        ax.set_xlabel("Température (°C)")
        ax.set_ylabel("Résistivité (Ω·m)")
        ax.set_title("Résultats Expérimentaux :\n Résistivité du disque d'aluminium en fonction de la température")
        ax.legend(loc='lower right')
        plt.savefig("Expérience1.png")
        plt.tight_layout()
```

Python code used to process the results – Z-score

```
# Constantes
rho0 = 28e-8    # Pour l'aluminium
T0 = 20
alpha0 = 4.03e-3

rho1 = 1.2e-7  # Pour l'acier
alpha1 = 4.0e-3

def Z_score():
    Z = []
    for i in range(len(X)):
        sigma = Yerr[i]
        delta = abs(Y[i] - (rho0 * (1 + alpha0 * (X[i] - T0))))
        Z.append(delta / sigma)

    fig, ax = plt.subplots()
    ax.scatter(X, Z, marker='+', label="Z-score")
    ax.set_xlabel("Température (°C)")
    ax.set_ylabel("Z-Score")
    ax.set_title("Estimation de la compatibilité avec le modèle pour l'
```

Python code used to simulate the dissipated power as a function of braking

```
def Comparaison_profils_freinage():
        Y_exp = np.array([23.8, 24.2, 25.6, 26.5, 24.3, 26.2, 27.5, 28.3,
                                        32.5, 33.9, 34.3, 35.5, 34.9, 36.2, 38.1, 38.9,
                                        39.6, 38.8, 39.9, 39.9, 40.1])
        X_exp = np.linspace(0, 1200, len(Y_exp))

        v0 = 5000
        tau_v = 400
        t_f = 1200

        # Profils de vitesse
        v_const = [v0] * len(X_exp)
        v_lin = [v0 * (1 - t / t_f) for t in X_exp]
        v_exp = [v0 * np.exp(-t / tau_v) for t in X_exp]

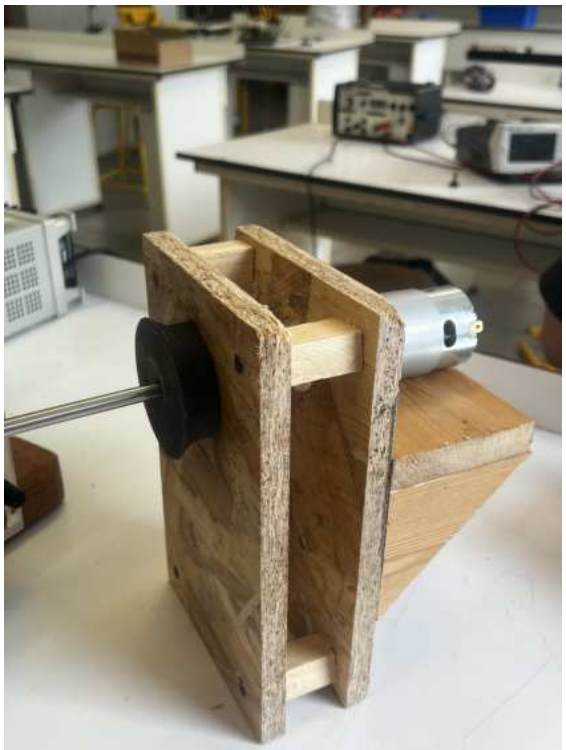
        # Températures théoriques pour chaque profil
        def T_model(v_list):
                return [alpha / (h * S) * v_list[i]**2 * (1 - np.exp(-X_exp[i] / tau)) + Tair for i in range(len(X_exp))]

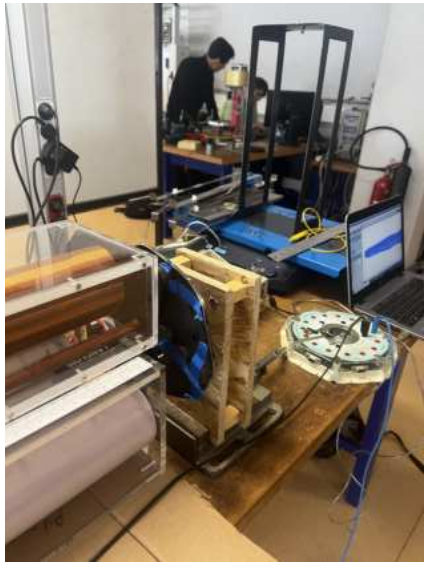
        T_const = T_model(v_const)
        T_lin = T_model(v_lin)
        T_exp = T_model(v_exp)

        # Dérivées et puissances
        def puissance(Y_model):
                dT = [(Y_model[i+1] - Y_model[i]) / (X_exp[i+1] - X_exp[i]) for i in range(len(Y_model)-1)]
                return [C * dT[i] + h * S * (Y_model[i] - Tair) for i in range(len(dT))]

        P_const = puissance(T_const)
        P_lin = puissance(T_lin)
        P_exp = puissance(T_exp)
```

Photos of the different setups







Notation used

- $\vec{F_L}$: Lorentz force
- P_{total}: power dissipated by the system
- ω: angular velocity of the disc
- R: radius of the disc
- B: amplitude of the magnetic field
- δQ: elementary heat
- τ: characteristic time of the heating
- ρ_0: resistivity at T_0
- α_0: temperature coefficient at T_0
- T_0: reference temperature